PCBC H Parametrics Practice Quiz: Answer Key

12 points

Every value below was checked in Python (sympy and 30-digit arithmetic) before typesetting. Work is on the left, the reason for each line is on the right.

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1. $x = \sqrt{t},\ y = 2t - 3,$ for t in $[0,4]$ *Non-Calculator*

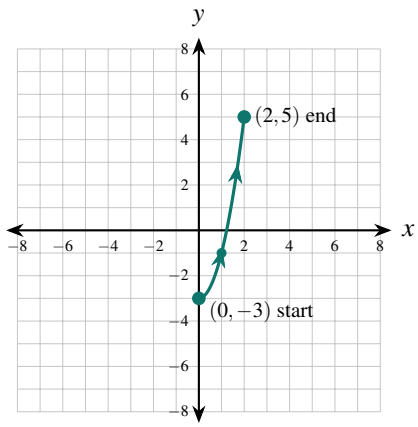
$\boxed{y = 2x^2 - 3, \quad 0 \le x \le 2}$

Graph: right half of the parabola from $(0,-3)$ to $(2,5)$, moving up and to the right.

Work	Why
$x = \sqrt{t}$	Pick the equation that gives t in one step.
$x^2 = t$	Square both sides. Note $x = \sqrt{t} \ge 0$ already.
$y = 2(x^2) - 3$	Substitute $t = x^2$ into y.
$y = 2x^2 - 3$	Simplify.
$t = 0 \Rightarrow x = 0$	$x = \sqrt{t}$ only increases on $[0,4]$, so the smallest t gives the smallest x.
$t = 4 \Rightarrow x = 2$	and the largest t gives the largest x. (Check y: -3 to 5, matches.)
$\boxed{y = 2x^2 - 3,\ 0 \le x \le 2}$	State the restriction with the equation.

t	(x,y)
0	$(0,-3)$
1	$(1,-1)$
4	$(2,5)$

Both endpoints plus one point in between is enough here, since $x = \sqrt{t}$ never turns around. Closed interval, so solid dots at both ends and arrows along the curve, not at the ends.



Practice scoring (3 pts): 1 for the equation, 1 for $0 \le x \le 2$, 1 for the graph with endpoint dots and direction arrows.

Trap: drawing the whole parabola. $x = \sqrt{t}$ can never be negative, so the left half does not exist, and t stops at 4, so the curve stops at $(2,5)$.

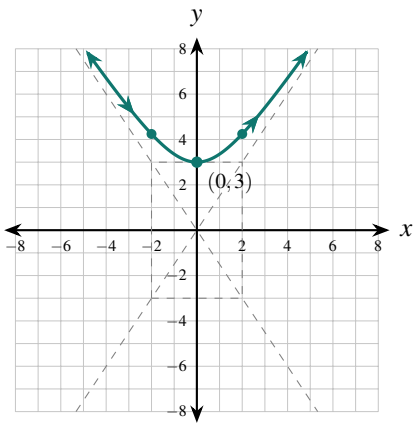
2. $x = 2\tan t,\ y = 3\sec t,$ for t in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ *Non-Calculator*

$\boxed{\frac{y^2}{9} - \frac{x^2}{4} = 1,\ \text{upper branch only: } y = \sqrt{\frac{9}{4}x^2 + 9}\ \ (y \geq 3)}$
x in $(-\infty, \infty)$. Traced left to right: down to $(0, 3)$ at $t = 0$, then up.

Work	Why
$\tan t = \frac{x}{2}, \quad \sec t = \frac{y}{3}$	Isolate each trig function.
$\sec^2 t - \tan^2 t = 1$	Pythagorean identity ($\tan^2 t + 1 = \sec^2 t$).
$\left(\frac{y}{3}\right)^2 - \left(\frac{x}{2}\right)^2 = 1$	Substitute.
$\frac{y^2}{9} - \frac{x^2}{4} = 1$	Hyperbola, center $(0, 0)$, opens up and down.
$\sec t > 0$ on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$	Cosine is positive there, so $y = 3\sec t$ is positive.
$y^2 = 9 + \frac{9}{4}x^2$	Solve for y.
$y = +\sqrt{\frac{9}{4}x^2 + 9}$	Positive root only: top branch, lowest point $(0, 3)$.
$y = \pm\frac{3}{2}x$	Asymptotes, slope $\pm\frac{a}{b} = \pm\frac{3}{2}$. Dashed guides only.

t	(x, y)
$-\frac{\pi}{4}$	$(-2, 3\sqrt{2})$
0	$(0, 3)$
$\frac{\pi}{4}$	$(2, 3\sqrt{2})$

$3\sqrt{2} \approx 4.24$ for plotting. As t goes up, $\tan t$ goes from $-\infty$ to ∞, so x runs left to right; y falls to 3 at $t = 0$, then rises. The excluded endpoints $t = \pm\frac{\pi}{2}$ are not finite points ($\tan t \to \pm\infty$), so the curve continues: arrowheads at the ends. Arrows along the curve show direction.



mid-curve arrows: direction
end arrowheads: curve continues

Practice scoring (3 pts): 1 for $\frac{y^2}{9} - \frac{x^2}{4} = 1$, 1 for keeping only the top branch (positive root), 1 for the graph with direction and asymptote guides.

Trap: drawing both branches. On this interval $\sec t$ is never negative, so the bottom branch is never traced. Compare $x = 4\cot t,\ y = 3\csc t$ on $(0, \pi)$ from the day 2 worksheet: also upper branch only, but traced right to left.

3. $y = x^2 + 6x + 5$ *Non-Calculator*

$\boxed{x = t,\ y = t^2 + 6t + 5,\ \text{for } t \text{ in } (-\infty, \infty)}$ and $\boxed{x = t - 3,\ y = t^2 - 4,\ \text{for } t \text{ in } (-\infty, \infty)}$

Work	**Why**
Representation 1: $x = t$	Simplest choice: let x be the parameter.
$y = t^2 + 6t + 5,\ \ t \text{ in } (-\infty, \infty)$	Replace every x with t. The parabola has every real x, so t is every real number.
Representation 2: $y = (x^2 + 6x + 9) - 9 + 5$	Complete the square: half of 6 is 3, $3^2 = 9$.
$y = (x+3)^2 - 4$	Vertex form, vertex $(-3, -4)$.
$t = x + 3 \ \Rightarrow\ x = t - 3$	Let t be the inside of the square.
$y = t^2 - 4,\ \ t \text{ in } (-\infty, \infty)$	Substitute.
Check: $(t-3)^2 + 6(t-3) + 5 = t^2 - 6t + 9 + 6t - 18 + 5 = t^2 - 4$	Plug $x = t - 3$ into the original to confirm.

Practice scoring (3 pts): 1 for each correct pair of equations, 1 for stating the parameter interval on both. Any other correct pair also counts (for example $x = t + 1,\ y = (t+4)^2 - 4$).

Trap: leaving off "for t in $(-\infty, \infty)$". The two equations and the interval are a package deal. Also watch the sign: $t = x + 3$ means $x = t - 3$, not $t + 3$.

4. Football punted from $h_0 = 2.5$ ft at $\theta = 40^\circ$ with $v_0 = 80$ ft/sec. *Calculator*

$\boxed{x = (80\cos 40^\circ)t, \quad y = -16t^2 + (80\sin 40^\circ)t + 2.5}$
In flight $\boxed{\approx 3.262 \text{ sec}}$, travels $\boxed{\approx 199.897 \text{ ft}}$ horizontally.

Work	Why
$x = (80\cos 40^\circ)t \approx 61.284t$	Horizontal: $v_0 \cos\theta$ times t.
$y = -16t^2 + (80\sin 40^\circ)t + 2.5 \approx -16t^2 + 51.423t + 2.5$	Vertical: gravity term, $v_0 \sin\theta$ times t, plus starting height.
$0 = -16t^2 + (80\sin 40^\circ)t + 2.5$	The ball lands when $y = 0$.
$t = \dfrac{-80\sin 40^\circ \pm \sqrt{(80\sin 40^\circ)^2 - 4(-16)(2.5)}}{2(-16)}$	Quadratic formula with $a = -16$, $b = 80\sin 40^\circ$, $c = 2.5$.
$t = \dfrac{-51.423 \pm \sqrt{2804.326}}{-32}$	$b^2 = 2644.326$, and $-4ac = +160$.
$t \approx 3.262$ or $t \approx -0.048$	Reject the negative time.
$x = (80\cos 40^\circ)(3.26184\ldots)$	Plug the unrounded t into x (keep it stored in the calculator).
$x \approx 199.897$ ft	Round only at the end.

Calculator setup (TI-Nspire), and a graph check

- Angle mode: **Degree**, or type the $^\circ$ symbol after 40. In radian mode $\sin 40$ means 40 radians and every number is wrong.
- Quadratic: in a Calculator page, type the formula above with $80\sin(40^\circ)$ in place of b, or store b first so nothing gets rounded.
- Graph check: Graphs page, Menu, Graph Entry/Edit, Parametric. Enter $x1(t) = 80\cos(40^\circ)t$ and $y1(t) = -16t^2 + 80\sin(40^\circ)t + 2.5$ with $0 \le t \le 3.5$. Window x: 0 to 220, y: -5 to 50. Trace (Menu, Trace, Graph Trace) and type t-values near 3.262: y crosses 0 there with $x \approx 199.9$.
- Not asked, but quick to see on the same trace: the top of the path is at $t = \frac{80\sin 40^\circ}{32} \approx 1.607$ sec, height ≈ 43.818 ft.

Practice scoring (3 pts): 1 for both parametric equations, 1 for the flight time, 1 for the horizontal distance.

Trap: rounding t to 3.26 before finding x gives 199.784, which is off at the thousandths. The day 2 worksheet's supplied answer for the archer problem (240 ft/sec, 30°, 3 ft) lists 1559 ft for the same reason: t was rounded to 7.5 first. Full precision gives $t \approx 7.525$ and $x \approx 1564.025$ ft.